Florian Julé

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Engineer with 8+ years developing complex electromechanical systems and currently completing an M.S. in Robotics. Experienced in product development, systems integration, robotics software (C++/Python/ROS 2), and hardware development from concept through production.

Projects

Driver-Assisted Stair-Climbing Wheelchair — Apr 2026 – Present

- Conceived and developed a 4WD articulated wheelchair for stair-climbing and off-road mobility, spanning mechanical design, ROS 2 software architecture (C++/Python), perception, and system integration on Jetson Orin Nano with a Gazebo digital twin.
- Architected and implemented the perception system from scratch across two cameras: an Intel RealSense D435i running YOLO26 semantic segmentation (trained on ADE20K) for general navigation and obstacle/stair detection, plus a D405 depth camera dedicated to stair navigation.

Cooperative Payload Transport with a Drone Swarm — Jan – Mar 2026

- Drove trajectory generation for a drone swarm end to end: formulated and solved a CasADi OCP for collision-free, dynamically feasible paths, then deployed and validated the full pipeline on real hardware via ROS 2 and Crazyswarm2.

EKF SLAM on a TurtleBot3 — Jan – Mar 2026

- Built a full SLAM stack from scratch in C++/ROS 2: 2D geometry library, simulator, odometry, and Extended Kalman Filter SLAM with unknown data association using LiDAR landmark detection.

Object Manipulation with Franka Arm — Nov – Dec 2025

- Led perception and system integration for a 4-person team, delivering YOLO-based pick-and-place software for a Franka robotic arm on real hardware.

Work Experience

Joby Aviation — **Santa Cruz, CA**

Staff Mechanical Engineer – Special Projects, Flight Research — Nov 2023 – Feb 2025

- Owned mechanical system design, integration and test across novel aircraft programs in Joby's flight research division, spanning retractable landing gear, structural components, and cross-system integration.

Tail and Effectors IPT Lead Engineer – S4-2.1 production aircraft — Oct 2018 – Nov 2023

- Cross-functional team lead for the development of tail, tilt nacelles, control surfaces, and propellers, advancing airframe designs towards FAA certification and production; managed 5 direct reports.
- Held end-to-end delivery responsibility for the first production-prototype aircraft and its conforming drawing package.
- Granted patent for the inboard nacelle tilt mechanism: Rotor assembly deployment mechanism.

Tilt Nacelles Lead Engineer – S4-2.0, pre-production aircraft — Oct 2017 – Oct 2018

- Owned design, analysis, build, and test for the tilt nacelles on the Joby S4 pre-production aircraft.
- Served as flight test engineer throughout the flight test program.

Mechanical Engineer – S4-1.0, first full-scale eVTOL aircraft — Feb 2017 – Oct 2017

- Supported composites manufacturing and structural testing for the first Joby S4 full-scale prototype.

Education

Northwestern University — **Evanston, IL**
M.S. in Robotics — Sep 2025 – Sep 2026

University of Michigan — **Ann Arbor, MI**
M.S. in Aerospace Engineering — Sep 2015 – Dec 2016

Arts & Métiers ParisTech — **Metz/Paris, France**
B.S. in Mechanical & Industrial Engineering — Sep 2013 – Jun 2015

Skills

Robotics	Sensor Integration (IMU, Cameras, LiDAR, Encoders), Localization (Kalman & Particle Filters), SLAM, Motion Planning, Control (PID, Feedforward)
Software	Python, C++, ROS 2, Git, Linux
Mechanical	Mechanism design, Composite structure, CAD (Catia, Onshape), FEA (Abaqus), GD&T (ASME Y14.5)
Languages	French, English